MECHENG 726
Question/Answer Booklet

THE UNIVERSITY OF AUCKLAND

SEMESTER TWO 2025
Campus: City

MECHANICAL ENGINEERING
Acoustics for Engineers
(Time allowed: THREE hours)

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

NOTE: Answer **ALL** questions.
This exam comprises 5 questions worth 20 marks each.
Calculators permitted.
This is an open book exam.

READ THESE INSTRUCTIONS BEFORE COMMENCING

1. Write your name and ID number on this page.
2. Write neatly and legibly. Use dark ink suitable for reproduction. Pencil for diagrams is acceptable.
3. The work justifying your answer should be written in the space provided directly after each question.
4. Cross out neatly any notes or other work you do not wish to have assessed. **DO NOT REMOVE ANY PAGES FROM ANY ANSWER BOOK.**
5. After the examination:
   a. Check that you have filled in your details on the front page correctly;
   b. Personally hand the books to the supervisor. **YOU MUST NOT REMOVE ANY ANSWER BOOKS FROM THE EXAM ROOM.**

1. A time-harmonic plane wave has a complex pressure field given by

$$p_i(x, y, t) = A \exp\{\mathrm{i}\omega t - \mathrm{i}kx \cos\beta + \mathrm{i}ky \sin\beta\},$$

where $A = 1$ Pa, $\beta = 45°$ and $\omega = 2000\pi$ rad/s. The wave propagates through a still atmosphere with a speed of sound of 340 m/s and density of 1.2 kg/m^3. This wave is incident on a rigid plane surface located at $y = 0$ and the reflection of the incident wave from this surface produces a reflected wave with a complex pressure field given by

$$p_r(x, y, t) = A \exp\{\mathrm{i}\omega t - \mathrm{i}kx \cos\beta - \mathrm{i}ky \sin\beta\}.$$

The total complex pressure field is the sum of the incident and reflected pressure fields. i.e. $p = p_i + p_r$.

(a) Calculate the sound pressure level of the incident wave (with complex pressure p_i).

(2 marks)

Use the space below for any working required for Question 1(a).

(b) Determine the sound pressure level of the total field on the rigid plane surface at $y = 0$.

(2 marks)

Use the space below for any working required for Question 1(b).

(c) Calculate the maximum and minimum sound pressure levels above the rigid plane surface and the first (lowest) locations above the $y = 0$ plane at which these occur.

(6 marks)

Use the space below for any working required for Question 1(c).

(d) Derive expressions for u and v for the total acoustic field.

Hint: The linearised Euler equations relate the acoustic pressure to the acoustic particle velocity in the x-direction, u, and the acoustic particle velocity in the y-direction, v, and are given by

$$\rho_0 \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x}, \qquad \rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p}{\partial y}.$$

(4 marks)

Use the space below for any working required for Question 1(d).

(e) Determine the acoustic particle velocity on the rigid plane surface located at $y = 0$. Comment on whether your result is expected.

(2 marks)

Use the space below for any working required for Question 1(e).

(f) Derive an expression for the time-average acoustic intensity in the x-direction. Also, calculate the sound intensity level for the time-average acoustic intensity in the x-direction.

Hint: the time-average acoustic intensity in the x-direction can be calculated using the following formula:

$$I = \frac{1}{2}\Re\{pu^*\},$$

where $\Re\{\cdot\}$ denotes that the real value of the argument should be taken, p is the complex pressure of the total acoustic field and u^* is the complex conjugate of the complex acoustic particle velocity in the x-direction of the total acoustic field.

(4 marks)

Use the space below for any working required for Question 1(f).

2.

(a) The spectrum of a steady noise is measured and found to contain only sound in 6 one-third octave bands. The level in each band is listed in the table below.

1/3 octave band centre frequency (Hz)	1/3 octave band sound pressure level (dB)
200	64
250	66
315	68
400	63
500	64
630	65

(i) Determine the sound pressure levels for the 250 Hz and 500 Hz octave bands.

(4 marks)

Write your answer to Question 2(a)(i) in the space below.

(ii) Calculate the overall sound pressure level of the noise.

(2 marks)

Write your answer to Question 2(a)(ii) in the space below.

(iii) Would you expect the overall A-weighted sound pressure level to be higher or lower than the unweighted overall sound pressure level calculated in Question 2(a)(ii)? Please explain your answer.

(2 marks)

Write your answer to Question 2(a)(iii) in the space below.

(b) A factory worker works a four-hour shift and has two different activities during that time separated by a 15-minute break. The following $L_{A,eq}$ values were measured for the 3 periods.

Time period	$L_{A,eq}$ (dB)
8:00 am – 10:00 am	87
10:00 am – 10:15 am	50
10:15 am – 12:00 noon	84

(i) What is the equivalent continuous A-weighted sound pressure level for the worker for the entire 4-hour shift?

(4 marks)

Write your answer to Question 2(b)(i) in the space below.

(ii) What is the Noise Dose of the worker for this shift?

(4 marks)

Write your answer to Question 2(b)(ii) in the space below.

(c) The noise limits specified in the Auckland Unitary Plan use the quantities $L_{A,eq,15\ min}$ and $L_{A,F,max}$. Describe what these two quantities are.

(2 marks)

Write your answer to Question 2(c) in the space below.

(d) New Zealand workplace law requires that a worker should not be exposed to a noise event which has an L_{peak} which exceeds 140 dB. A worker is exposed to a noise event with a peak acoustic pressure of 150 Pa. Does this event contravene New Zealand law?

(2 marks)

Write your answer to Question 2(d) in the space below.

3.

(a) A monopole source produces 1 Watt of sound power. The sound has a single frequency component at 3 kHz. The speed of sound is 340 m/s and the air density is 1.2 kg/m^3.

(i) The complex acoustic pressure produced by this source is given by

$$p = \frac{B}{r} \exp\left\{\mathrm{i}\omega\left(t - \frac{r}{c_0}\right)\right\}$$

Define each of the terms which appear in this equation and state their SI units.

(3 marks)

Write your answer to Question 3(a)(i) in the space below.

(ii) Determine the magnitude of B in the equation for the complex acoustic pressure produced by the source.

(4 marks)

Write your answer to Question 3(a)(ii) in the space below.

(iii) Estimate the sound pressure level 3 m from the source if the sound radiates into an anechoic environment.

(2 marks)

Write your answer to Question 3(a)(iii) in the space below.

(iv) Estimate the sound pressure level 3 m horizontally from the source if the source is placed 2 m above a rigid ground surface.

(5 marks)

Write your answer to Question 3(a)(iv) in the space below.

Continue your answer to Question 3(a)(iv) in the space below.

(b) Two incoherent monopole sources each produce 1 Watt of sound power. The sound radiates into an anechoic environment.

(i) Estimate the sound pressure level at a location which is 3 m from both sources.

(2 marks)

Write your answer to Question 3(b)(i) in the space below.

(ii) Estimate the sound pressure level at a location which is 3 m from one source and 5 m from the other source.

(4 marks)

Write your answer to Question 3(b)(ii) in the space below.

4. You have been asked to provide an acoustic assessment for a fabrication company. The fabrication plant room has a reverberant acoustic field with a sound pressure level of 85 dB. The noise is the result of one CNC machine which operates continuously and produces noise at a constant level. The plant room is modelled as a single Sabine room which has a reverberation time T_{60} = 1 s and volume 200 m^3.

   (a) Determine the likely reverberant field sound pressure level if 1 more identical CNC machine is added to the plant room. Assume that both machines produce the same sound power and that the noise from both machines is uncorrelated.

   (4 marks)

   Write your answer to Question 4(a) in the space below.

(b) Calculate the minimum sound absorption area required to ensure that the reverberant sound pressure level within the plant room is below 85 dB when two machines are operating.

(4 marks)

Write your answer to Question 4(b) in the space below.

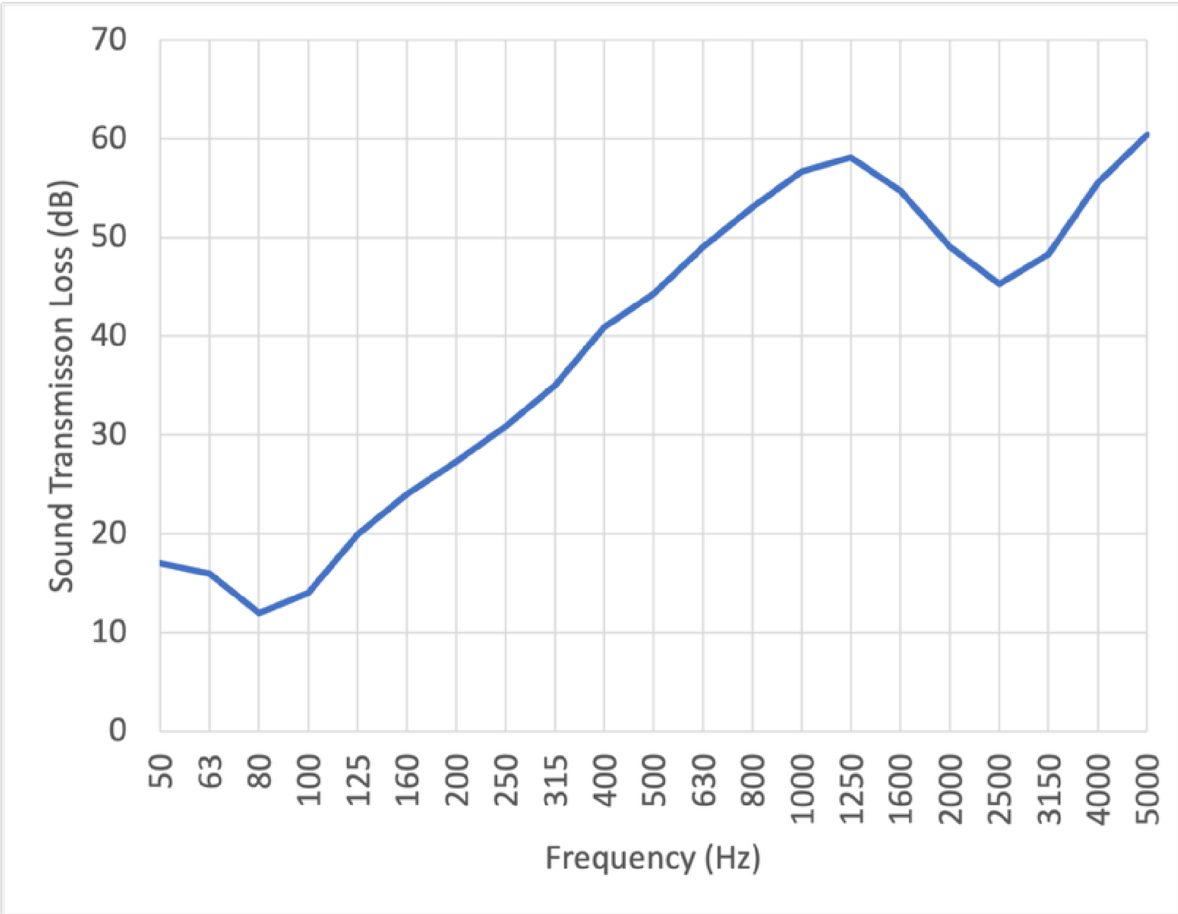
(c) A control room is situated adjacent to the plant room with volume 40 m^3 and reverberation time T_{60} = 0.5 s. A 10 m^2 wall with unknown construction separates the plant and control rooms. Calculate the required sound reduction index of the wall if the noise level in the plant room is 85 dB, but the level in the control room must not exceed 45 dB.

(5 marks)

Write your answer to Question 4(c) in the space below.

(d) The wall system separating the two rooms considered in Question 4(c) is built and tested at the Acoustics Testing Service at the University of Auckland. The construction of the wall is a double leaf partition consisting of a 90 mm wooden stud with one layer of 13 mm gypsum plasterboard on either side with no batts.

The results from this test are shown in the figure below. Identify the mass-air-mass resonance frequency and the critical frequency.



(3 marks)

Write your answer to Question 4(d) in the space below.

(e) The client is not satisfied with the sound transmission loss provided by the wall considered in Question 4(d). Suggest two solutions that will improve the sound transmission loss performance of the wall system and explain how these solutions achieve this.

(4 marks)

Write your answer to Question 4(e) in the space below.

5. You have been sent to commission a new recording studio in Auckland. There are two rooms identical in shape being rectangular prisms with internal dimensions of 5 m wide × 4 m long × 3.5 m high. The reverberation time in both rooms is 0.3 s.

   (a) Calculate the total sound absorption area for each studio room.

   (3 marks)

   Write your answer to Question 5(a) in the space below.

   (b) Define the Schroeder frequency and calculate its value for one of the studio rooms.

   (2 marks)

   Write your answer to Question 5(b) in the space below.

(c) It is found that there is a room mode present which produces a peak in the sound pressure level at low frequencies at certain locations within the studio rooms.

List two absorption methods to target acoustic energy in a narrow low frequency band.

(2 marks)

Write your answer to Question 5(c) in the space below.

(d) The second studio room is situated directly beneath the first studio room with a simple floor-ceiling system separating the two rooms. The maximum expected reverberant sound pressure level in each studio room is 90 dB.

Calculate the sound reduction index of the floor-ceiling system required to ensure that the reverberant sound pressure level in the other studio room due to sound transmission is less than 20 dB.

(3 marks)

Write your answer to Question 5(d) in the space below.

(e) The floor-ceiling system considered in Question 5(d) is a simple double-leaf system, with one leaf/panel being 20 mm thick and the other leaf/panel being 30 mm thick. Both leaves/panels are made from a homogeneous, isotropic material with a Young's modulus of 2×10^9 Pa, a Poisson's ratio of 0.3 and a density of 700 kg/m^3. The panels are separated by a 90 mm air gap with no batts.

(i) Calculate the mass-air-mass resonance frequency of the floor-ceiling system.

(2 marks)

Write your answer to Question 5(e)(i) in the space below.

(ii) Calculate the critical frequency of each of the leaves/panels.

(2 marks)

Write your answer to Question 5(e)(ii) in the space below.

(f) To ensure the floor-ceiling system is adequate for impact noise, the floor-ceiling system is tested at the Acoustic Testing Service at the University of Auckland. Define the Normalized Impact Sound Pressure Level and describe the method used for measuring it.

(2 marks)

Write your answer to Question 5(f) in the space below.

(g) The record label running the studio is worried about impact noise from the drum kit propagating from the upper studio room to the lower studio room.

(i) Suggest two different practical measures to improve the impact sound insulation of the floor and briefly explain the underlying physical principles.

(2 marks)

Write your answer to Question 5(g)(i) in the space below.

(ii) Define ‘flanking transmission’ and discuss how it may occur in this scenario.

(2 marks)

Write your answer to Question 5(g)(ii) in the space below.

Extra paper for answering questions. Please make sure you indicate the number of the question being answered.

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